MIL-F-18801(PA)
27 January 1975
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

FUZE, IMPACT M1123 AND M1125

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for two types of impact fuzes entitled Fuze, Impact M1123 and M1125 which are used in the Sergeant warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-E-5272	**- Environmental Testing, Aeronautical and Associated Equipment, General Specifications for.**

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169(MU)	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460	**- Soldering of Electrical connections and Printed Wiring Assemblies, Procedure For.**

FSC: 1135

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:57Z
Check the source to verify that this is the current version before use.

DRAWINGS

ORDNANCE CORPS

8830094 - Fuze Impact M1123.
8830101 - Body.
8830670 - Fuze Impact M1125.
8838216 - Connector Assembly.
8838221 - Converter Assembly.
8842013 - Tester, Output Voltage, Impact Fuze, XM1123 and XM1125.
8858016 - M1123 and M1125 Impact Fuzes.
EL-8854155 - Equipment List.

PUBLICATIONS

PICATINNY ARSENAL

QAI-PA-1 - Quality Assurance Instructions for Reporting Data on Nuclear Weapons.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The Impact Fuze shall comply with all requirements specified on the drawing(dwg) 8830094 and 8830670, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A six letter serial code shall be assigned to the item. Serial codes required on these items shall be obtained by means of a letter addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SARPA-QA-T-M. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial codes required (unused code numbers shall be returned).

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:57Z
Check the source to verify that this is the current version before use.

3.5 Coded serialization requirements (see 6.1.2). The contractor shall supply feedback data on punch cards for any item for which coding is required in section 3. The feedback data shall be in accordance with QAI-PA-1.

3.6 Insulation resistance. The insulation resistance between the inner connector pin and the connector shall be 20 megohms minimum at 500 + 50 volts direct current (VDC) when tested as specified in 4.4.1.

3.7 Capacitance. The capacitance between the inner connector pin and the connector shell of the Impact Fuze shall be 105 + 15 micromicrofarads at 1000 + Hertz (Hz) when tested as specified in 4.4.2.

From	To	Capacitance Requirement	Test Code No.
Inner Connector Pin	Connector Shell	105 + 15 micromicrofarads	040001

3.8 Output voltage. The output voltage of the Impact Fuze upon being subjected to a shock of 450 + 50 gravity units (g) shall be 90 volts minimum and positive with respect to the fuze case when tested as specified in 4.4.3.

Drop	Voltage Requirement	Test Code Nos.
1	90 volt, minimum	050001
2	90 volt, minimum	050002
3	90 volt, minimum	050003

3.9 Environmental testing. The Impact Fuze shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The Impact Fuze shall also comply with 4.3.3.1 before and after each environmental test. Testing shall be as specified in 4.4.4. Tests shall be conducted in the order specified.

3.9.1 Transportation vibration. The Impact Fuze shall withstand vibration for 15 minutes in each of the three mutually perpendicular axes shown in Figure 1 at the frequencies and amplitudes as specified below without evidence of damage or affecting function when tested as specified in 4.4.4.1. The frequency shall be varied from 8 to 500 to 8 Hz in 15 minutes. Upon completion of the vibrations, the Impact Fuze shall comply with the requirements of 3.6, 3.7 and 3.8.

Frequency Range (Hz)	Time Cycle	Constant acceleration or Double Amplitude
8-55-8	7.5 minutes	0.060 + 0.006 inch
55-500-55	7.5 minutes	10 + 1 g

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:57Z
Check the source to verify that this is the current version before use.

3.9.2 Operational vibration. The Impact Fuze shall withstand a programmed sequence of band-limited white Gaussian noise and combined noise and sine wave vibration without evidence of damage or affecting function when tested as specified in 4.4.4.2. The total time of test shall be 210 seconds; 70 seconds in each of three mutually perpendicular axes, as shown in Figure 1. The complex vibration test shall consist of the following five sections programmed continuously:

a. White Gaussian noise at a level of 15.4 gravity units (g) root mean square (RMS) band-limited between 15 and 1500 Hz for 1 second.
b. A 2.8 g RMS swept sine wave superimposed on 5.25 g RMS of white Gaussian noise band-limited between 15 and 1500 Hz for 15 seconds. The sine wave shall be swept from 800 to 1300 Hz at a rate increasing directly with the frequency, one such sweep taking the full 15 seconds.
c. A 5.6 g RMS swept sine wave superimposed on 10.5 g RMS of white Gaussian noise band-limited between 15 and 1500 Hz for 15 seconds. The sine wave shall be swept from 1300 to 800 Hz at a rate decreasing directly with frequency, one such sweep taking the full 15 seconds.
d. A 2.8 g RMS sine wave superimposed on 5.25 g RMS of white Gaussian noise band-limited between 15 and 1500 Hz for 38 seconds. The sine wave shall be swept from 15 to 1500 cps and back to 15 Hz at a rate increasing with frequency. One such cycle shall take the full 38 seconds.
e. White Gaussian noise at a level of 15.4 g RMS band-limited between 15 and 1500 Hz for 1 second.

The output voltage of the fuze shall not exceed 32 volts durint vibration when measured as specified in Figure 4, with the exception that the total capacitance as measured from TNC connector shall equal 1200 ± 20 micromicrofarad with the cable connected to the scope without the Impact Fuze in the circuit. After the test the Impact Fuze shall comply with the requirements of 3.6, 3.7, and 3.8.

3.9.3 High temperature. The Impact Fuze shall show no evidence of damage or deterioration, such as distortion, cracks, or swelling, when subjected to 160 ± 5 degrees Fahrenheit (°F.) as specified in 4.4.4.3 and shall meet the requirements of 3.8 while at that temperature. After return to ambient temperature, the Impact Fuze shall comply with the requirements of 3.6, 3.7, and 3.8.

3.9.4 Low temperature. The Impact Fuze shall show no evidence of damage or deterioration, such as distortion, cracks, or contracting when subjected to -65 ± 5°F. as

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:57Z
Check the source to verify that this is the current version before use.

specified in 4.4.4.4, and shall meet the requirements of 3.8 while at that temperature. After return to ambient temperature, the Impact Fuze shall comply with the requirements of 3.6, 3.7, and 3.8.

3.9.5 Humidity. The Impact Fuze shall withstand relativy humidity of 95 + 5 percent at a temperature of 160 + 5°F. for three consecutive 24-hour test cycles without evidence of damage or affecting function when tested as specified in 4.4.4.5. Within one hour after conclusion of the test, the Impact Fuze shall comply with the requirements of 3.6, 3.7, and 3.8.

3.10 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.11 Test codes. The contractor shall provide data feedback on punch cards in conformance with Quality Assurance Instruction QAI-PA-1. The data required to be coded, and the code numbers, are specified in the applicable paragraphs in this section.

3.12 Workmanship.

3.12.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.12.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.12.3 Burrs. Parts shall be free from burrs.

3.12.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.12.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contractor requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 8830094 and 8830670) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

Source: https://assist.dla.mil Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests shall be conducted in order specified in Table I.

Table I

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Insulation resistance (see 3.6)		X
3. Capacitance (see 3.7)		X
4. Output voltage (see 3.8)		X
5. Transportation vibration (see 3.9.1)	X	
6. Operational vibration (see 3.9.2)	X	
7. High temperature (see 3.9.3)		X
8. Low temperature (see 3.9.4)		X
9. Humidity (see 3.9.5)	X	

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:57Z
Check the source to verify that this is the current version before use.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

Source: https://assist.dla.mil -- Downloaded: 202640-26103:57Z
Check the source to verify that this is the current version before use.

Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Fuze Impact, M1123 and M1125 (see dwg 8830094 and 8830670) prior to assembly.

Categories	Characteristic	Method of inspection
Critical:		
1.	Presence of nib in bottom of C bore (see dwg 8830101).....	Manual/Visual
Major:		
101.	Improper soldering (see dwg 8838221 or 8838216).....	Visual
102.	Improper cleaning (see dwg 8858016).....	Visual
Minor:		
201.	Foreign matter present other than on indicated surface.....	Visual
202.	Evidence of poor workmanship (see 3.12).....	Visual

4.3.2.1.2 Fuze, Impact M1123 and M1125 (see dwg 8830094 and 8830670) during assembly.

Categories	Characteristics	Method of inspection
Critical:		
1.	Orientation of ceramic elements.....	Visual
2.	Orientation of spring.....	Visual
3.	Improper cleaning and sealing procedure (dwg 8858016).....	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

Major: None defined.

Minor: None defined.

4.3.2.1.3 Fuze, Impact M1123 and M1125 (see dwg 8830094 and 8830670) following assembly.

Categories	Characteristics	Method of inspection
Critical:	None defined.	
Major:		
101.	Mislocation of mounting hole	Gage
102.	Improper torque	Manual/Visual
103.	Objective evidence relative to material missing or improper (see 3.1)	Visual
104.	Evidence of poor workmanship (see 3.12)	Visual
Minor:		
201.	Improper identification	Visual
202.	Cap missing	Visual
203.	Serialization improper or missing	Visual

4.3.2.1.4 Preparation for delivery (see 5).

Categories	Characteristics	Method of inspection
Critical:	None defined.	
Major:		
101.	Packing material and packing not proper or missing	Visual
102.	Marking not in accordance with MIL-STD-1169(MU)	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.12)	Visual

4.3.3 Testing. All tests shall be performed. Tests sahall be performed in the order specified below.

Test	Test Classification
a. Insulation resistance	Major
b. Capacitance	Major
c. Output voltage	Major
d. Operational vibration	Major
e. Humidity	Major

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

Of these tests d shall be considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Insulation resistance, capacitance, and output voltage. Each Impact Fuze shall be tested in accordance with 4.4.1 through 4.4.3. If any item fails to comply with 3.6 through 3.8, the item shall be rejected and removed from the lot.

4.3.3.2 Operational vibration, and humidity. A sample shall be selected at random in accordance with Table IV. The sample shall be tested in accordance with 4.4.4.2 and 4.4.4.5. If any item fails to comply with 3.9.2 and 3.9.5, the lot shall be rejected.

Table IV

Sample sizes for destructive testing

Number Required to be Delivered	Sample Size for No Failures
2-8	1
9-180	2
181-500	3
501-800	5
801-1300	7

Lot Size Required = Number Required to be Delivered + Sample Size.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (See EL-8854155). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Insulation resistance. Connect a megohmmeter or approved test circuit device between the inner connector pin and the connector shell with the positive lead connected to the center pin and the negative lead to its connector shell. Apply 500 ± 50 VDC to the fuze and measure the resistance. In the test circuit, protective resistance shall be applied to limit the current to 5 milliamperes under a short circuit condition.

4.4.2 Capacitance. The circuit shown in Figure 2, or its approved equivalent, shall be connected to the Impact Fuze connector, and the resulting capacitance measured. The capacitance of the test cable shall be subtracted from the capacitance of the cable-Impact Fuze combination and the resultant difference shall be recorded as the capacitance of the Impact Fuze. Record values (see 3.11).

4.4.3 Output voltage. The Impact Fuze shall be mounted on a fixture (see dwg 8842013), or approved equivalent. The assembly shall be mounted with the cover "UP" - i.e., away from the mounting surface of the striking rod and shall be secured to the rod with the approved clamping device. The fuze shall be dropped in such a manner that the impact (small) end of the rod will strike the plate normal to the plate surface. A releasing device shall be used to produce the same pre-fall conditions for each drop. The output voltage shall be measured as specified in Figure 3, i.e., voltage read on the scope multiplied by two is the output voltage of the Impact Fuze and the output voltage measuring circuit shall have an impedance consisting of two megohms ± 3 percent in parallel with a capacitance of 450 ± 20 micromicrofarads which includes the capacitance of the measuring instrument and test cable but excludes the Impact Fuze. The fuze in the fixture shall be dropped from a height which will subject the fuze to 450 ± 50 g upon impact. Each fuze shall be dropped three times and the output voltage shall be recorded for each drop (see 3.11).

4.4.4 Environmental testing (see 3.9).

4.4.4.1 Transportation vibration. The Impact Fuze shall be rigidly mounted to a fixture capable of being supported in each of three mutually perpendicular axes, with the fixture mounted on a suitable shake table. Vibrate for 15 minutes in each of three mutually perpendicular axes, as shown in Figure 1, at the frequencies and amplitudes specified in 3.9.1. Total test time shall be 45 minutes. After the completion of the test, the Impact Fuze shall be tested for compliance with 3.6, 3.7, and 3.8.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

4.4.4.2 Operational vibration. The Impact Fuze shall be rigidly mounted to a fixture capable of being supported in each of three mutually perpendicular axes, with the fixture mounted on a suitable shake table. Connect the Impact Fuze to the test circuit as shown in Figure 3 except that the total capacitance as measured from TNC connector shall be 1200 + 20 micromicrofarads which includes capacitance of the measuring instrument and test cable but excludes the Impact Fuze. Monitor the output voltage during vibration. Vibrate the fuze in three mutually perpendicular axes, as shown in Figure 1, to meet the requirements of 3.9.2 using a standard noise or a complex wave tape having the characteristics listed in Table V.

Table V

Time (Sec)	Type of Signal	Ratio of Test Level to Calibration Level	RMS g Level
0-20	Noise (Calibration) (1)	0.7	Set gain to give 1.4 g RMS
20-30	None	-	-
30-31	Noise (1)	7.7	15.4
31-46	Noise (1) plus sinusoid (2)	2.97	5.95
	Noise alone	2.62	5.25
	Sinusoid alone	1.40	2.8
46-61	Noise (1) plus sinusoid (3)	5.95	11.9
	Noise alone	5.25	10.5
	Sinusoid alone	2.80	5.6
61-99	Noise (1) plus sinusoid (4)	2.97	5.95
	Noise alone	2.62	5.25
	Sinusoid alone	1.40	2.8
99-100	Noise (1)	7.7	15.4

(1) Band-limited (15-1500 Hz) white Gaussian noise.
(2) Sine wave swept from 800 to 1300 Hz at a rate increasing directly with frequency. One full sweep in 15 seconds.
(3) Sine wave swept from 1300 to 800 Hz at a rate decreasing directly with frequency. One full sweep in 15 seconds.
(4) Sine wave swept from 15 to 1500 Hz and back to 15 Hz at a rate increasing directly with frequency. One full cycle in 38 seconds.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:57Z
Check the source to verify that this is the current version before use.

For creating the random noise, the output of a General Radio random noise generator, or its equivalent, may be used if proper care is taken to insure the correct amplitude distribution of the signal. For testing the noise generator or any other associated equipment, the use of an amplitude distribution analyzer is suggested. For insuring the correct noise bandwidth, a filter with an asymptatic slope of at least 24 decibel (DB) per octave and 3 DB points at 15 and 1500 Hz shall be considered acceptable. Following the vibration, the Impact Fuze shall be tested for the requirements of 3.6, 3.7, and 3.8.

4.4.4.2.1 Method and procedure with the standard noise tape. Before applying the tape, the transfer function or frequency response of the entire vibration system shall be as flat as is consistent with the present state of the art. Plus or minus 3 DB or better over the band from 15 to 1500 Hz shall be considered as an objective in this respect. After proper equalization of the system the standard noise tape is applied. The overall gain of the shake system is then adjusted so that with the noise calibration signal, the g level listed in 4.4.4.2 is obtained. With the same gain setting, the tape is run for the remainder of the test and the g levels obtained will conform to the levels listed in Table V.

4.4.4.3 High temperature. The Impact Fuze shall be placed in a test chamber. The chamber's temperature shall be raised and stabilized at 160 ± 5°F. (see 6.4). After the fuze has been subjected to this temperature, it shall be tested for compliance with the requirement of 3.6 (see 6.5). Allow the fuze to return to room temperature without forced cooling, either by fan or refrigerated air. When the fuze is at ambient temperature, test for compliance with the requirements of 3.6, 3.7, and 3.8.

4.4.4.4 Low temperature. The Impact Fuze shall be placed in a test chamber. The chamber's temperature shall be lowered and stabilized at -65 ± 5°F. (see 6.4). After the fuze has been subjected to this temperature for a minimum of 4 hours and while still at this temperature, it shall be tested for compliance with the requirement of 3.6 (see 6.5). Allow the fuze to return to room temperature without forced warming. When the fuze is at ambient temperature, test for compliance with the requirements of 3.6, 3.7, and 3.8.

4.4.4.5 Humidity. The Impact Fuze with cap shall be placed within the test chamber and shall be subjected to the humidity test cycles as specified in MIL-E-5272, Paragraph 4.4, Procedure 1. All temperatures shall be maintained at ± 5°F. Three such 24 hour cycles shall be performed consecutively. The pH values shall be adjusted by the use of chemically pure hydrochloric acid or sodium hydroxide. Upon completion of the last cycle, the Impact Fuze shall be removed from the test chamber. The excess moisture shall be wiped from the fuze surface and the fuze shall be tested for compliance with the requirements of 3.6, 3.7, and 3.8.

Source: https://assist.dla.mil Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The Impact Fuze shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The Impact Fuze shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The Impact Fuze shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 3.5). Data specified in 3.11 and 4.3.4.1 will be delivered as identified on a DD Form 1664 when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Measurement in extreme environments. The output voltage test fixture may be operated at room temperature providing the output voltage reading is observed within 15 seconds after removal from the environmental conditioning chamber.

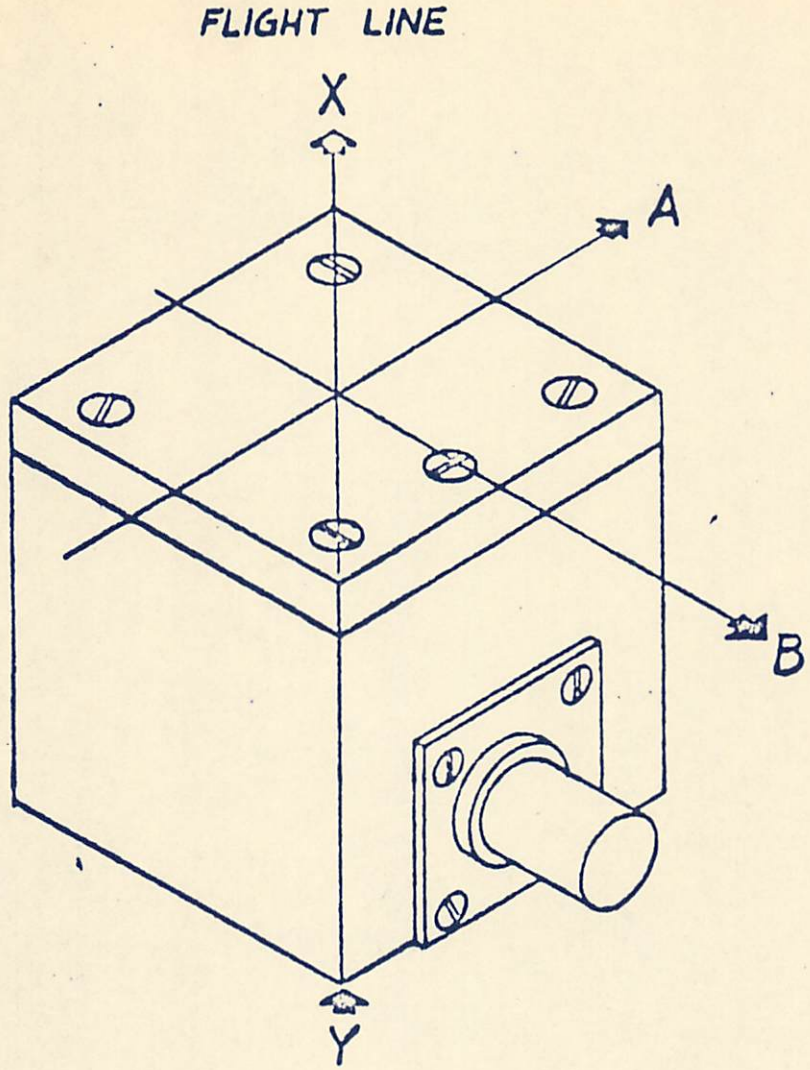
6.6 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-2066 (Rev. 1) dated 10 August 1965.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1135-A179

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

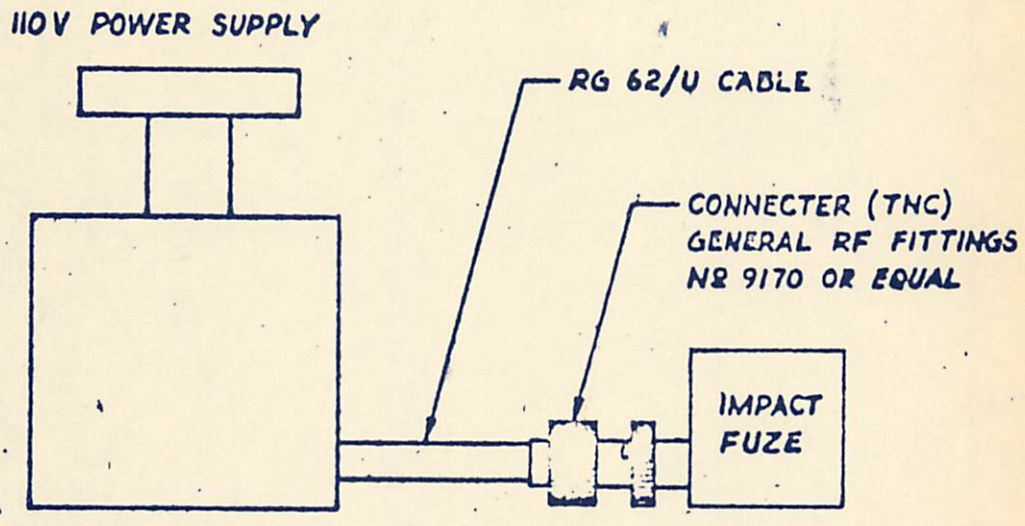


M1123 or M1125 Impact Fuze

Figure 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

TEST CIRCUIT CAPACITANCE TEST



NOTE:

For true capacitance of the impact fuze subtract capacitance of cable from bridge reading RG 62/U - 13.6 μμf/foot

EQUIPMENT

Bridge - Electro measurements Capacitance Bridge model 270-H Electric measurements incorporated or approved equal.

Figure 2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

TESTER, OUTPUT VOLTAGE,

IMPACT FUZE

ASSIGNED DRAWING NO. 8842013

FOR

XM1123 and XM1125

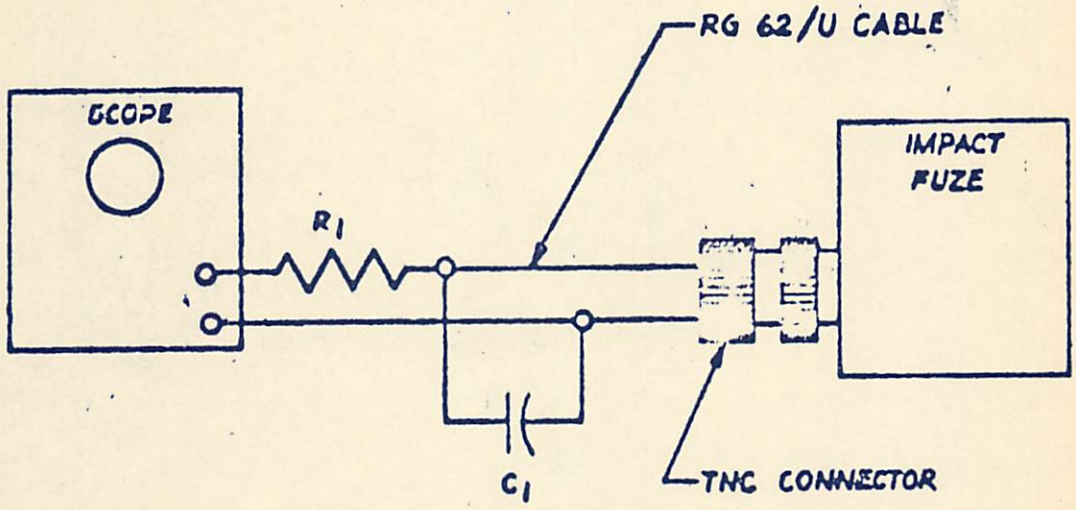
FUZE, IMPACT

NOTE: THE INCLUSION OF DESIGN AND DESCRIPTIVE ENGINEERING INFORMATION IN THIS PURCHASE DESCRIPTION TENTATIVELY OMITTED PENDING APPROVAL FROM QUALITY ASSURANCE DIVISION OF PICATINNY ARSENAL.

Figure 3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:57Z
Check the source to verify that this is the current version before use.

CIRCUIT FOR OUTPUT VOLTAGE TEST



(1) R1 = R Scope (matched within 1%).
(2) R1 + R Scope = 2 megohms ± 3%.
(3) C1 Value to be so chosen that total capacitance as measured from TNC connector equals 450 uufd. ± 20 uuf with cable with cable connected to Scc
(4) Connect R1 & C1 to Scope with leads as short as possible.
(5) Calibrate scope with DC supply accurate to 1%.
(6) The voltage measured on the scope in one-half the total impact fuze voltage output.

Figure 4.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:57Z
Check the source to verify that this is the current version before use.